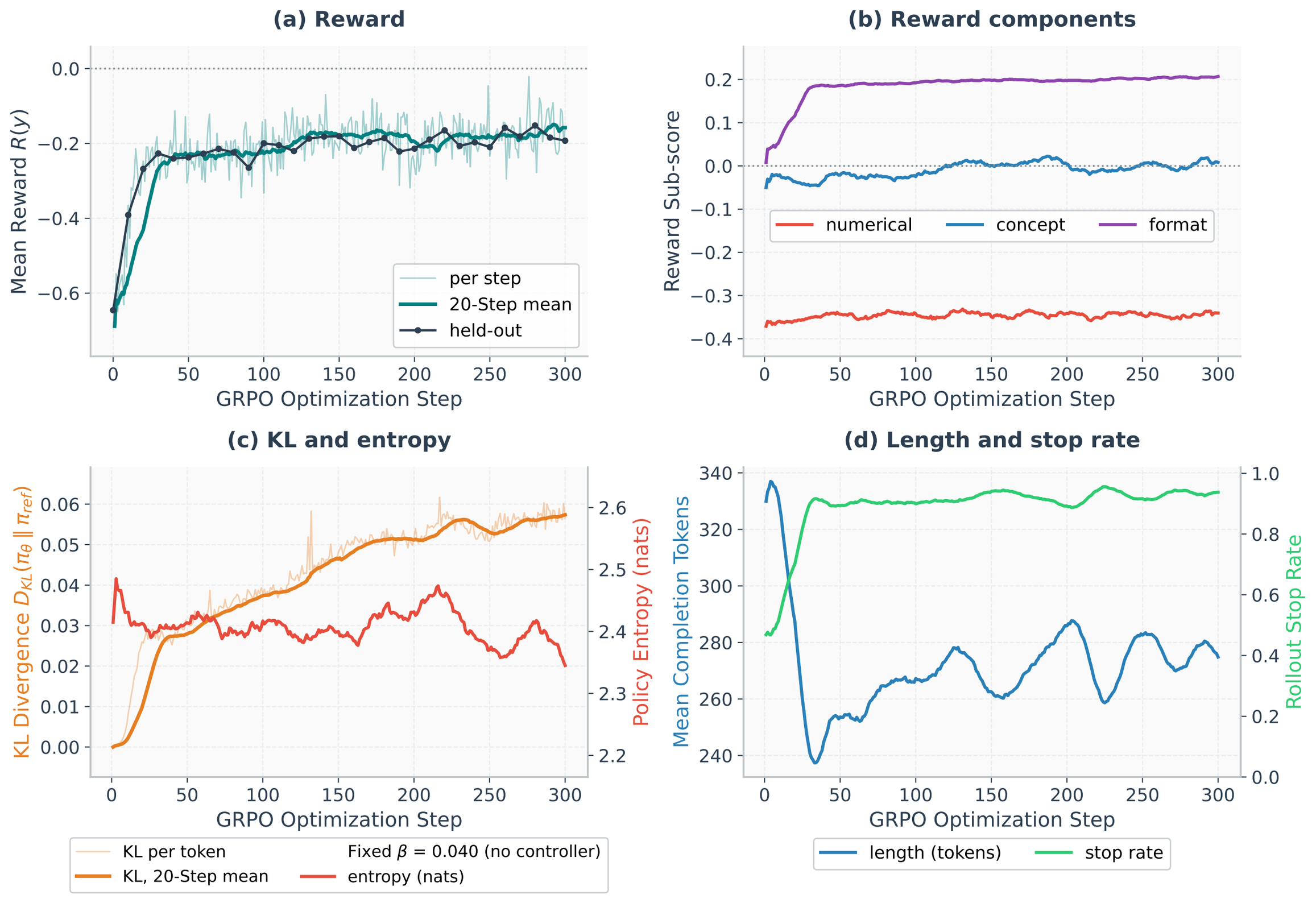

(a) Reward
Mean Reward $R(y)$
GRPO Optimization Step
per step
20-Step mean
held-out
(b) Reward components
Reward Sub-score
numerical
concept
format
GRPO Optimization Step
(c) KL and entropy
KL Divergence $D_{KL}(\pi_\theta \| \pi_{ref})$
Policy Entropy (nats)
GRPO Optimization Step
KL per token
KL, 20-Step mean
Fixed β = 0.040 (no controller)
entropy (nats)
(d) Length and stop rate
Mean Completion Tokens
Rollout Stop Rate
GRPO Optimization Step
length (tokens)
stop rate